Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Cubes and Cuboids: Counting Faces, Edges and Vertices

Some shapes are solid: they have length, breadth and height, so they take up space. These are called **three-dimensional** shapes or **solid shapes**. A cube of sugar, a drink can and a match box are all solid shapes.

To describe a solid shape we talk about its **faces**, its **edges** and its **vertices** (one is a *vertex*).

- A **face** (or surface) is a flat or curved side of the shape.
- An **edge** is a line where two faces meet. It may be straight or curved.
- A **vertex** is a point corner where edges meet.

In this section you will learn the properties of a **cube** and a **cuboid** — that is, how many faces, edges and vertices each one has.

A **cube** is the solid shape you know from a cube of sugar. All of its faces are squares, and they are all the same size.

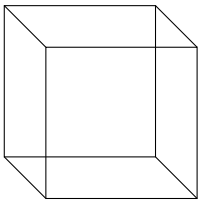
Look at fig-01, which shows a cube drawn as a see-through solid. Count its parts and you will find these properties:

1. It is a solid shape.
2. It has **6 square faces** — every face is a square.
3. It has **12 edges**.
4. It has **8 vertices**.

A **cuboid** is a solid shape like a match box. Its faces are rectangles, and opposite faces are the same size. Fig-02 shows a cuboid, with leader lines naming its **face**, one of its **vertices** and one of its **edges**. A cuboid has these properties:

1. It has **6 faces**.
2. It has **12 edges**.
3. It has **8 vertices**.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

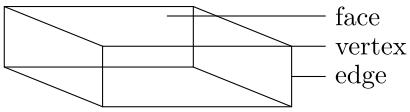
Worked example

Look at fig-01, the cube. Let us count its properties step by step.

- **Faces.** A face is a flat side. The cube has a top, a bottom, and four sides around it — that is $1 + 1 + 4 = \mathbf{6}$ square faces.
- **Edges.** An edge is a line where two faces meet. Each of the two squares in fig-01 has four sides, and the four corners are joined, so the cube has $4 + 4 + 4 = \mathbf{12}$ **edges**.
- **Vertices.** A vertex is a point corner where edges meet. The two squares in fig-01 have four corners each, so the cube has $4 + 4 = \mathbf{8}$ **vertices**.

Now look at fig-02, the cuboid. It has the same number of parts: **6** faces (all rectangles), **12** edges and **8** vertices. The only difference is that a cube's faces are all squares, while a cuboid's faces are rectangles.

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

Practice

1. How many vertices has a cuboid?
2. How many edges has a cube?
3. How many faces has a cube, and what shape are they?
4. On a sheet of paper, draw a cube and a cuboid the way fig-01 and fig-02 are drawn. On each one, label one face, one edge and one vertex, and write how many faces, edges and vertices the shape has.

Section 2: Pyramids and Cones: Counting Faces, Edges and Vertices

A pyramid is named after the shape of its base. Look at fig-03, fig-04, fig-05 and fig-06: the triangular base pyramid (fig-03), the square base pyramid (fig-04), the pentagonal based pyramid (fig-05) and the rectangular based pyramid (fig-06). Each name tells you the shape of the base. A triangular base pyramid is also called a tetrahedron.

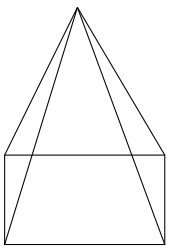
Every pyramid has one flat base and sloping triangular faces that all meet at a single point on top, called the apex. To describe a pyramid you count three things:

- its faces — the flat surfaces (the base plus the sloping triangles);
- its edges — the lines where two faces meet;
- its vertices — the corner points where edges meet.

The table below lists these for the four pyramids in the figures:

Shape	Faces	Edges	Vertices
Triangular based Pyramid (tetrahedron)	4	6	4
Square based Pyramid	5	8	5
Pentagonal base Pyramid	6	10	6
Rectangular based Pyramid	5	8	5

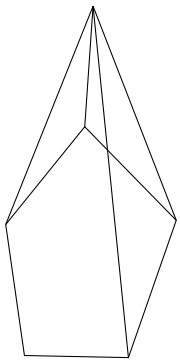
Visual



Rectangular based Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

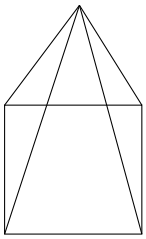
Visual



Pentagonal based Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

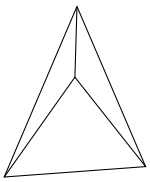
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

Visual



Triangular base Pyramid (Tetrahedron)

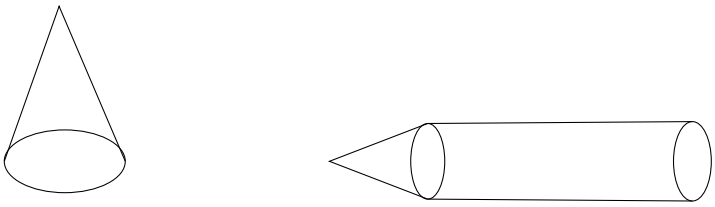
A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

A cone is a different solid. Look at fig-07, where the cone is drawn twice — once standing upright and once lying on its side. It does not have a set of flat triangular faces like a pyramid. Instead:

1. It has two faces — one curved face and one flat face (the circular base).
2. It has one edge — the rim where the curved face meets the flat base.
3. It has one vertex — the pointed apex at the top.

So a cone is a solid with a single curved face, a single flat face, one edge and one vertex.

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

Worked example. How many faces, edges and vertices does a square base pyramid have?

The name tells you the base is a square (fig-04). Reading across the table, a square base pyramid has 5 faces (1 square base + 4 sloping triangles), 8 edges and 5 vertices.

Practice

1. How many faces, edges and vertices does a pentagonal base pyramid have?
2. A tetrahedron has 4 faces and 6 edges. How many vertices does it have?
3. How many faces, edges and vertices does a cone have?
4. How many edges and vertices does a rectangular base pyramid have?

Check your work

Attempt every question before you read this page.

Cubes and Cuboids: Counting Faces, Edges and Vertices

1. 8 vertices

   A cuboid has two squares of corners: the nearer four and the farther four. $4 + 4 = 8$.
2. 12 edges

   Each square has 4 sides, and the four corresponding corners are joined, so the cube has $4 + 4 + 4 = 12$ edges.
3. 6 square faces.

   A cube has a top, a bottom and four sides: $1 + 1 + 4 = 6$ faces. Every face is a square, and all the squares are the same size.
4. Cube: 6 faces, 12 edges, 8 vertices. Cuboid: 6 faces, 12 edges, 8 vertices.

   A face is a flat side, an edge is a line where two faces meet, and a vertex is a point corner where edges meet. Check that the cube you drew has every face a square, and that you have labelled one face, one edge and one vertex on each drawing.

Pyramids and Cones: Counting Faces, Edges and Vertices

1. 6 faces, 10 edges, 6 vertices.

   Look across the row for the pentagonal base pyramid in the table: it has 6 faces, 10 edges and 6 vertices.
2. A tetrahedron (a triangular base pyramid) has 4 vertices.

A tetrahedron is the same as a triangular base pyramid. The table gives it 4 faces, 6 edges and 4 vertices, so it has 4 vertices.

3. A cone has 2 faces (one curved and one flat), 1 edge and 1 vertex.
4. A rectangular base pyramid has 8 edges and 5 vertices.

   Like the square base pyramid, the rectangular base pyramid (fig-06) has 8 edges and 5 vertices.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-4cc9b098	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____